

Synthetic Methods | Hot Paper |

Palladium-Catalyzed Zinc-Amide-Mediated C–H Arylation of Fluoroarenes and Heteroarenes with Aryl Sulfides

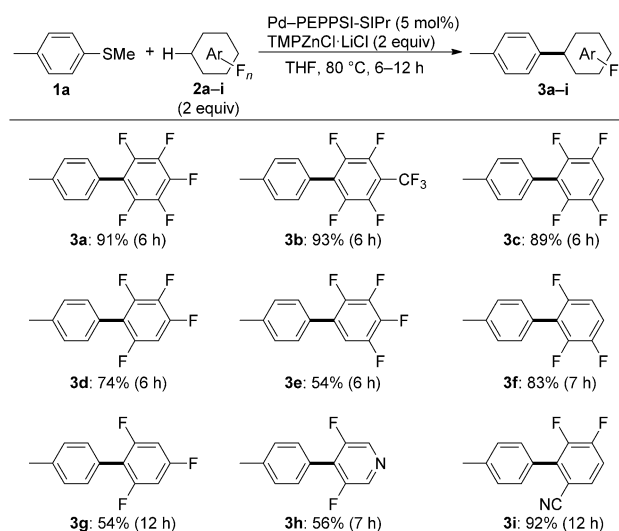
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Abstract: C–H arylation of polyfluoroarenes and heteroarenes with aryl sulfides proceeds smoothly with the aid of a palladium–N-heterocyclic carbene catalyst. A bulky zinc amide, TMPZnCl·LiCl, plays a key role as an effective base to generate the corresponding arylzinc species in situ. This arylation protocol is practically much easier to perform than our previous method, which necessitates preparation of the arylzinc reagents in advance from the corresponding aryl halides. Aryl sulfides that are prepared through sulfur-specific reactions, such as S_NAr sulfanylation and extended Pummerer reactions, undergo this direct arylation, offering interesting transformations that are otherwise difficult to achieve with conventional halogen-based organic synthesis.

Organosulfur compounds have played unique roles in organic chemistry owing to their intriguing reactivities, biological activities, or physical properties.^[1] Among them, aryl sulfides occupy an important position, and, therefore, new methods for the synthesis of aryl sulfides have been actively developed.^[2] In contrast, the development of new reactions of aryl sulfides has only been reported only scarcely. Transition-metal-catalyzed cross-coupling reactions of aryl sulfides^[3] are difficult to achieve because sulfur species can permanently poison the catalyst and C(sp²)–S bonds are rather strong and difficult to cleave. To overcome these difficulties, reactive Grignard reagents were employed as nucleophiles^[4] or aryl sulfides were used that have special structures that facilitate efficient conversion.^[5–7] We are interested in catalytic transformations of organosulfur compounds^[4e–g, 8] and recently developed general cross-coupling reactions of aryl sulfides with arylzinc reagents by using a palladium–NHC complex as catalyst.^[8b] Although our Negishi-type cross-coupling shows a wide scope and proceeds under mild conditions, the reactions necessitate preparation of the arylzinc reagents from the corresponding haloarenes in advance.

The transition-metal-catalyzed direct C–H arylation of arenes with aryl halides has emerged as a promising alternative to conventional cross-coupling reactions.^[9] However, due to the aforementioned difficulties in transforming aryl sulfides catalytically, only very recently Wang reported the first elegant example of a direct C–H arylation with aryl sulfides in which a palladium–diphosphine catalyst and sodium *tert*-butoxide were employed.^[10] Nevertheless, the arylation lacks generality: electron-neutral and -rich aryl sulfides reacted moderately, even at 140 °C, and only oxazole derivatives smoothly underwent the arylation. To expand the utility of aryl sulfides, we herein report efficient and general conditions for the C–H arylation of polyfluoroarenes^[11, 12] and heteroarenes^[13] with aryl sulfides.

The reaction of methyl *p*-tolyl sulfide (**1a**) with pentafluorobenzene (**2a**) was chosen as a model reaction. We screened palladium catalysts and bases (see the Supporting Information for optimization studies) to find that Pd–PEPPSI-SiPr ([1,3-bis(2,6-diisopropylphenyl)imidazolidin-2-ylidene](3-chloropyridine)palladium(II) dichloride^[14] and 2,2,6,6-tetramethylpiperidylzinc chloride–lithium chloride complex^[15, 16] (TMPZnCl·LiCl) are the best catalyst/base combination to afford **3a** in 91% yield (Scheme 1). As reported by us previously,^[8] Pd–PEPPSI-SiPr, a palladium–NHC complex, is again the key for success. TMPZnCl·LiCl is known as an exquisite base for deprotonative zincation of electron-deficient arenes or heteroarenes, tolerating various functional groups.^[15] We were delighted to find that TMPZnCl·LiCl generates a pentafluorophenylzinc species



Scheme 1. Scope of polyfluoroarenes.

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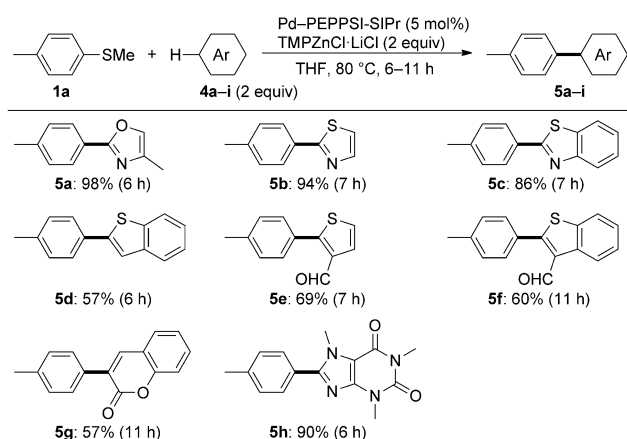
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in situ^[17] that is compatible with successive Negishi-type cross-coupling, which allowed us to achieve C–H arylation under mild conditions. Employment of other TMPZn bases, such as TMPZnCl·MgCl₂, TMP₂Zn·2MgCl₂, TMP₂Zn·2LiCl, or tBu₂(TMP)ZnLi,^[16b,c] led to much lower yields (see the Supporting Information). Another advantage of using the zinc base is that the resulting zinc methylthiolate species is not nucleophilic enough to induce conceivable S_NAr displacement of a fluoro group with a methylsulfanyl group.^[18]

A variety of polyfluoroarenes can be used for this reaction. Perfluorinated 2,3,5,6-tetrafluorobenzotrifluoride (**2b**) reacted smoothly to afford **3b** in excellent yield. For less fluorinated arenes, the efficiency of arylation is likely to depend heavily on the acidity of the proton that is deprotonated: reactions of 1,2,4,5-tetrafluoro-, 1,2,3,5-tetrafluoro-, and 1,2,4-trifluorobenzene gave the corresponding products **3c**, **3d**, and **3f** in high yields, whereas less acidic 1,2,3,4-tetrafluoro- and 1,3,5-trifluorobenzene were converted less efficiently to **3e** and **3g**, respectively.^[19] The arylation of **2f** took place regioselectively at the 4 position under the acidity control. 3,5-Difluoropyridine (**2h**) proved to be acidic enough to be arylated at the 4 position. A cyano group is also effectively electron-withdrawing to yield **3j** in excellent yield with exclusive regioselectivity.

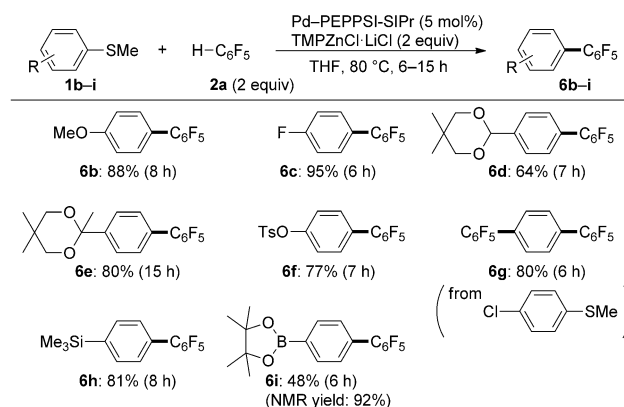
Not only polyfluoroarenes but also five-membered heteroarenes reacted under the reaction conditions (Scheme 2). Compared with Wang's procedure, 4-methyloxazole, thiazole, and



Scheme 2. Scope of heteroarenes.

benzothiazole reacted much more smoothly to give the 2-arylated products **5a–c** in high yields with exclusive regioselectivity. Besides azoles, thiophene derivatives reacted at the 2 position to yield **5d–f** in good yields. Six-membered coumarin and bioactive caffeine also gave the desired products **5g** and **5h**, respectively.

The scope of aryl sulfides is also broad (Scheme 3). Neither an electron-donating methoxy group nor an electron-withdrawing fluoro group retarded the reaction (**6b** and **6c**). Acetal-protected aldehydes and ketones were also compatible (**6d** and **6e**). A tosyloxy group, which is known to be a leaving group for cross-coupling reactions,^[8b,20] was tolerated to yield

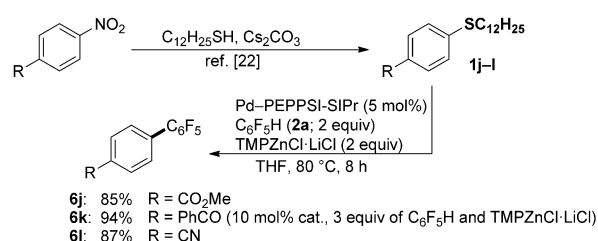


Scheme 3. Scope of aryl sulfides.

6f although 4-chlorothioanisole underwent twofold arylation to give terphenyl **6g**. Interestingly, a trimethylsilyl group and pinacoloboryl group survived under the basic conditions (**6h** and **6i**^[21]).

When combined with reactions that are unique to organosulfur compounds, this direct arylation offers interesting transformations that are otherwise difficult to achieve.

Aryl sulfides **1j–l** bearing a carbonyl or cyano group were synthesized by nucleophilic aromatic substitution of electron-deficient nitroarenes with dodecanethiol (Scheme 4).^[22] These

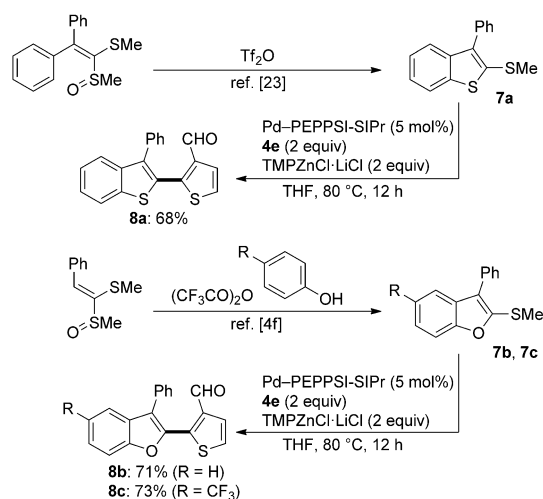


Scheme 4. S_NAr displacement of a NO₂ group followed by C–H arylation.

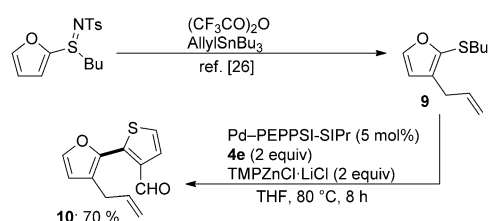
aryl sulfides underwent a catalytic pentafluorophenylation selectively at the dodecylsulfanyl unit with the polar functional groups untouched. The overall two-step transformation represents a displacement of a nitro group with a perfluoroaryl group by using the unique reactivity of the organosulfur species.

Our group previously reported the concise synthesis of multisubstituted benzothiophenes^[23] and benzofurans^[4f,24] containing a methylsulfanyl group through extended Pummerer chemistry of ketene dithioacetal monoxides^[25] (Scheme 5). We found that the thus synthesized compounds **7** are good substrates for the direct arylation with thiophene-3-carbaldehyde (**4e**) in spite of the steric hindrance of the neighboring phenyl group.

Padwa developed a 3-allylation of 2-heteroaryl sulfonylimines under Pummerer-type conditions.^[26] According to this protocol, we synthesized 2-butylsulfanyl-3-allylfuran (**9**), which was then



Scheme 5. Synthesis of 2,3-diarylbenzothiophene and benzofuran.

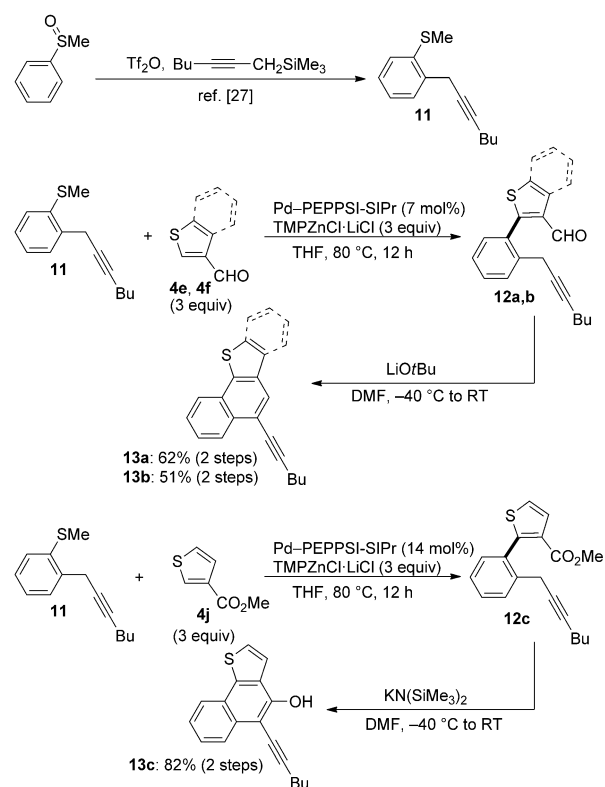


Scheme 6. 3-Allylation followed by 2-arylation.

subjected to the arylation conditions to yield the corresponding 3-allyl-2-thienylfuran **10** (Scheme 6).

Procter recently reported extended Pummerer reactions of aryl sulfoxides with propargylsilanes to provide 2-propargylaryl sulfides.^[27] We found that such a 2-propargylated product **11** is a useful precursor of naphthothiophene derivatives (Scheme 7). The arylation of thiophene derivatives **4e**, **4f**, and **4j** bearing a carbonyl group at the 3 position with **11** proceeded smoothly despite steric repulsion. We were glad to find that treatment of **12a** and **12b** with LiOtBu led to dehydration to yield alkynylated naphthothiophene **13a** and benzonaphthothiophene **13b**, respectively. Similarly, treatment of **12c** with KN(SiMe₃)₂ resulted in condensation followed by tautomerization to yield hydroxynaphthothiophene **13c** in high yield.

In summary, we have developed an efficient protocol for the direct arylation of polyfluoroarenes and heteroarenes with aryl sulfides. Our system is broad in scope and exhibits intriguing orthogonal reactivity, leaving the tosyloxy, trimethylsilyl, and pinacolatoboryl groups intact. This arylation can follow an S_NAr displacement of a nitro group with an alkylsulfanyl group or extended Pummerer reactions, which provides new transformations by taking advantage of the unique reactivities of organosulfur compounds. Our current studies are directed to disseminate C–S-based organic synthesis that complements C–X-based synthesis.



Scheme 7. Arylation of 3-carbonylthiophene derivatives with 2-propargyl-phenyl sulfide followed by condensation.

Experimental Section

The reaction of methyl *p*-tolyl sulfide (**1a**) and pentafluorobenzene (**2a**) is representative (Scheme 1, **3a**). Pd-PEPPSI-SIPr (17.0 mg, 0.025 mmol) was placed in a 20 mL Schlenk tube under nitrogen. After the addition of TMPZnCl·LiCl (1.0 mL, 1.0 mmol, 1.0 M in THF), pentafluorobenzene (168.1 mg, 1.0 mmol), methyl *p*-tolyl sulfide (69.1 mg, 0.50 mmol), and THF (1.0 mL), the resulting mixture was heated to 80 °C and stirred for 6 h at this temperature. After being cooled to 20 °C, the reaction was quenched with saturated NaHCO₃ aq. (20 mL) and the organic compounds were extracted with *n*-hexane (10 mL×3). The combined organic layer was passed through pads of anhydrous sodium sulfate and activated alumina, and concentrated in vacuo. Purification by chromatography on silica gel (*n*-hexane) provided **3a** as a white solid (117.1 mg, 0.45 mmol, 91%).

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Keywords: C–H arylation • C–S bond activation • organosulfur compounds • organozinc compounds • palladium

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